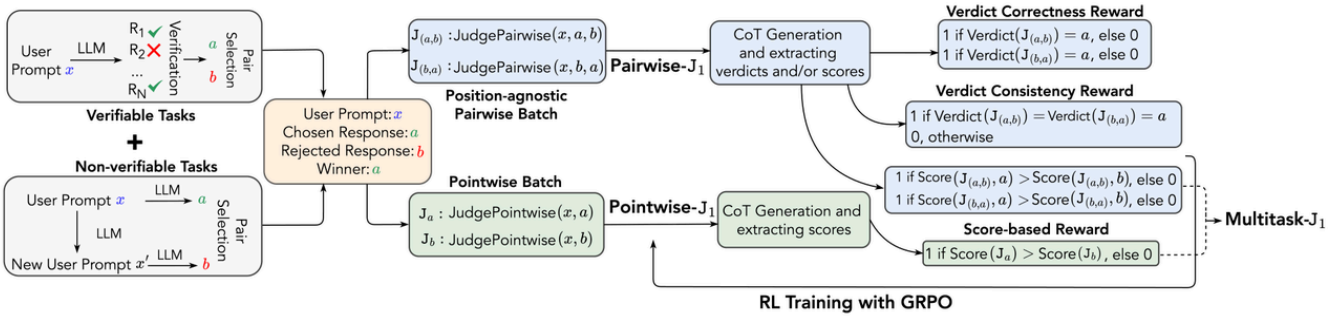
J1: INCENTIVIZING THINKING IN LLM-AS-A-JUDGE VIA REINFORCEMENT LEARNING

> #RewardModel #RL #Inference #CoT



- What problems does this framework solve?
  - Currently, it is common to use an LLM as the judge to be the reward model.
  - However, the precision of the rewarding heavily relies on the ability of the model. Also, even worse, LLM will have a hallucination that binds its decision with the position of the answer. That is, if the model tends to believe that the answer at index i of the answer array is the winner, it will always think in this way: even if the array is changed and the $i-th$ answer is actually bad.
  - This framework is to solve these problems.
- What does this framework do?
  - This framework teaches the reward model to think before deciding the winner plus to mitigate the position-binding problem via GRPO.
  - **Data Preparation**:
    - Use both mathematical problem sets and normal conversation sets.
    - For each prompt in the mathematical problem set, generate a pair of answers with one correct answer and one wrong answer.
    - For the conversation prompts, as they are non-verifiable, for each prompt x, we generate a correct response and a modified (noisy) version x' of x by modifying some details, so the answer to x' is not a proper answer to x. Then, use the answer of both x and x' to construct a pair.
  - **Reward Design**
    - Normal reward: 1 if the policy model successfully finds out the good answer; 0 otherwise.
    - Positional reward: 1 if the policy successfully finds out y_w for both (x, y_l, y_w) and (x, y_w, y_l) where the tuples have swapped position of the two answers.
  - **Training**
    - We use standard GRPO to finetune the model.
  - **Other Types of Outputs**
    - Pairwise J1 with Verdict, PaV: Given the input and the two answers, the model generates CoT and final decision.
    - Pairwise J1 with Scores, PaS: Given the input and the two answers, the model generates

CoT and the score of each answer.
        - Pairwise J1 with Scores & Verdict, PaVS: [PaV + PaS]
        - Pointwise J1, PoS: Evaluate each answer **one by one**. Separately provide a CoT+score without comparison.
        - Multitask Pairwise & Pointwise J1, MT: Train a model under both pair evaluation mode and the point evaluation mode.
- Experiment Results
    - Model Scale: The research team trained J1 models at 8B, 32B, and 70B scales based on Llama-3.1-Instruct and Qwen3.
    - SOTA Performance: J1 models achieved state-of-the-art (SOTA) performance across five major benchmarks including PPE, RewardBench, and JudgeBench.
    - Leapfrog Challenge: Most notably, the multi-task variant J1-Qwen-32B outperformed the massive 671B-parameter DeepSeek-R1 model, as well as the closed-source o1-mini and o3 models, on certain benchmarks despite being trained solely on synthetic data.
    - Qualitative Analysis: The model learned systematic evaluation strategies, including dynamically generating assessment criteria, composing reference answers, performing iterative self-correction, and providing improvement feedback for low-quality responses.
- Work Cited
    - Paper
    - Chat History